

ece4-exp

EC-Earth4 Experiment Configuration Made Simple

From hours of YAML editing to 30 seconds

The Problem

Creating an EC-Earth4 experiment config is painful:

- Edit 257-line YAML file manually
- Match components, grids, and processor counts
- Calculate node layouts and resource distribution
- Set platform paths, SLURM settings, forcing...
- Easy to make mistakes (typos, incompatible combinations)

Result: 2-4 hours for experienced users, longer for newcomers

The Solution

One command generates production-ready configs:

```
./ece4-exp generate \  
  --recipe gcm-sr.yml \  
  --sim-procs 1120 \  
  --expid my-experiment
```

That's it. You get:

- Correct component configuration
- Calculated node layout (10 nodes)
- Validated and ready to run

What You Get

Pre-tested recipes for common experiments:

- `gcm-sr.yml` - Coupled atmosphere-ocean
- `omip-sr.yml` - Ocean-only with ERA5 forcing
- `amip-sr.yml` - Atmosphere-only
- `ccycle-sr.yml` - With carbon cycle (LPJG)

Smart features:

- Auto-calculates processor distribution
- Fetches platform configs from EC-Earth4 repo
- Validates before you submit
- Works on BSC, ECMWF (to add a platform just a file with launcher templates is

Real Example: GCM vs OMIP

Coupled atmosphere-ocean (10 nodes):

```
./ece4-exp generate --recipe gcm-sr.yml --sim-procs 1120  
# OIFS + NEMO + XIOS + OASIS + RNFM
```

Ocean-only (2 nodes):

```
./ece4-exp generate --recipe omip-sr.yml --sim-procs 224  
# NEMO + XIOS (no atmosphere)
```

Different science, different resources - same simple command.

Setup Once, Use Forever

One-time setup:

```
git clone https://github.com/vlap/ece4-exp.git
cd ece4-exp
./setup.sh
./ece4-exp init-user
```

Then just:

```
./ece4-exp generate --recipe <type> --sim-procs <n>
```

Your platform, account, scratch are remembered.

Override when needed: `--platform ecmwf-hpc2020`

Get Started

Try it now:

```
git clone https://github.com/vlap/ece4-exp.git
cd ece4-exp
./QUICK_DEMO.sh
```

Documentation:

- `README.md` - Complete guide
- `DEMO.md` - Full walkthrough with examples
- `./ece4-exp --help` - Command reference

Compatible with: auto-ecearth4 (Autosubmit) and manual workflows

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